



DONGXING WASTE INCINERATION PLANT LEACHATE BIOGAS HARMLESS TREATMENT AND COMPREHENSIVE UTILIZATION PROJECT



Project title	Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project
Project ID	3851
Monitoring period	21-Jan-2022 to 31-Dec-2023
Original date of issue	22-Mar-2024
Most recent date of issue	19-Jun-2024
Version	03
VCS Standard Version	4.7
Prepared by	Yaya Xu, Henan BCCY Environmental Energy Co., Ltd.

CONTENTS

1	PROJECT DETAILS	4
1.1	Summary Description of the Implementation Status of the Project	4
1.2	Audit History.....	4
1.3	Sectoral Scope and Project Type	5
1.4	Project Proponent	5
1.5	Other Entities Involved in the Project	5
1.6	Project Start Date	5
1.7	Project Crediting Period	6
1.8	Project Location	6
1.9	Title and Reference of Methodology	8
1.10	Double Counting and Participation under Other GHG Programs	9
1.11	Double Claiming, Other Forms of Credit, and Scope 3 Emissions.....	9
1.12	Sustainable Development Contributions	9
1.13	Commercially Sensitive Information.....	12
2	SAFEGUARDS AND STAKEHOLDER ENGAGEMENT	12
2.1	Stakeholder Engagement and Consultation.....	12
2.2	Risks to Stakeholders and the Environment.....	17
2.3	Respect for Human Rights and Equity	21
2.4	Ecosystem Health	23
3	IMPLEMENTATION STATUS	24
3.1	Implementation Status of the Project Activity	24
3.2	Deviations	25
3.3	Grouped Projects	26
3.4	Baseline Reassessment.....	26
4	DATA AND PARAMETERS.....	26
4.1	Data and Parameters Available at Validation	26
4.2	Data and Parameters Monitored.....	31
4.3	Monitoring Plan.....	42
5	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	46
5.1	Baseline Emissions	46

5.2	Project Emissions	48
5.3	Leakage Emissions	50
5.4	GHG Emission Reductions and Carbon Dioxide Removals	50
APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION.....		55
APPENDIX 2: <SUPPORTING EVIDENCE>		56

1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project (hereafter referred to as the project) is located at Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P.R. China. The project is constructed and operated by Zhongmu Branch company of Henan BCCY Environmental Energy Co., Ltd.

Before the implementation of this project, Zhengzhou (East) environmental Protection Energy Project invested by Zhengzhou Dongxing Environmental Protection Energy Co., Ltd. incinerates domestic waste with a daily treatment capacity of 4,000t /d. Leachate will be produced during the storage and fermentation process in the garbage bin. The supporting treatment system of the factory is "pretreatment +UASB (upflow anaerobic sludge blanket) + secondary A/O-MBR aerobic system + nanofiltration NF+STRO" treatment process. A large amount of biogas is produced in the anaerobic treatment section of UASB.

This project collects the biogas generated by the anaerobic treatment section of leachate to generate electricity and connect to the grid, which can effectively utilize biogas resources.

The project combusts the leachate biogas to generate electricity and export it to the Central China Power Grid (CCPG). Total electricity supplied to the grid in the crediting period will be 118,470MWh. VCUs will be claimed from the methane destroyed and the grid electricity displaced. The estimated average GHG emission reductions of the project are 66,310tCO₂e and the total GHG emission reductions and removals are 663,100tCO₂e during the fixed 10 years crediting period.

The project started to construct on 15-09-2021 and started operation on 21-Jan-2022. This monitoring period is from 21-Jan-2022 to 31-Dec-2023. The total installed capacity is 3MW, including 3 sets with each capacity of 1MW. For this monitoring period, the net electricity delivered to CCPG was 20,740.137MWh. The total GHG emission reduction was 123,622 tCO₂e.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	21-Aug-2023	VCS	LGAI Technological Center, S.A. (Applus+ Certification)	/
Verification	21-Jan-2022 - 31-Dec-2023	VCS	LGAI Technological Center, S.A. (Applus+ Certification)	710 days

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	Scope 1: energy industries (renewable-/non-renewable sources) Scope 13: Waste handling and disposal
Project activity type	The project is neither a AFOLU project nor a grouped project

1.4 Project Proponent

Organization name	Henan BCCY Environmental Energy Co., Ltd.
Contact person	Lei Wang
Title	Project Manager
Address	8/F, Hengmei Business Building, 22 Dongfeng Road, Zhengzhou, China
Telephone	+86 371 56737901
Email	lwang@bccynewpower.com

1.5 Other Entities Involved in the Project

Organization name	Zhongmu Branch company of Henan BCCY Environmental Energy Co., Ltd
Role in the project	Project Operation Entity
Contact person	Yaya Xu
Title	Project Manager
Address	8/F, Hengmei Business Building, 22 Dongfeng Road, Zhengzhou, China
Telephone	+86 371 56737901
Email	xuyaya@bccynewpower.com

1.6 Project Start Date

Project start date	21-Jan-2022
--------------------	-------------

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

Justification	As per 3.8 of VCS Standard (Version 4.7), the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or removals. The project activity is a non-AFOLU project and started operation on 21-Jan-2022. Therefore, the project start date is 21-Jan-2022.
---------------	--

1.7 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	21- Jan -2022 to 20-Jan-2032

1.8 Project Location

The project is located in Dongxing Waste Incineration Plant, Zhengyoumo Village, Zhengang Town, Zhongmu County, Zhengzhou City, Henan Province P.R. China. The geographical coordinates of the project site are longitude 113.57° E and latitude 34.37° N. Figure 1.1 shows the location of the project.

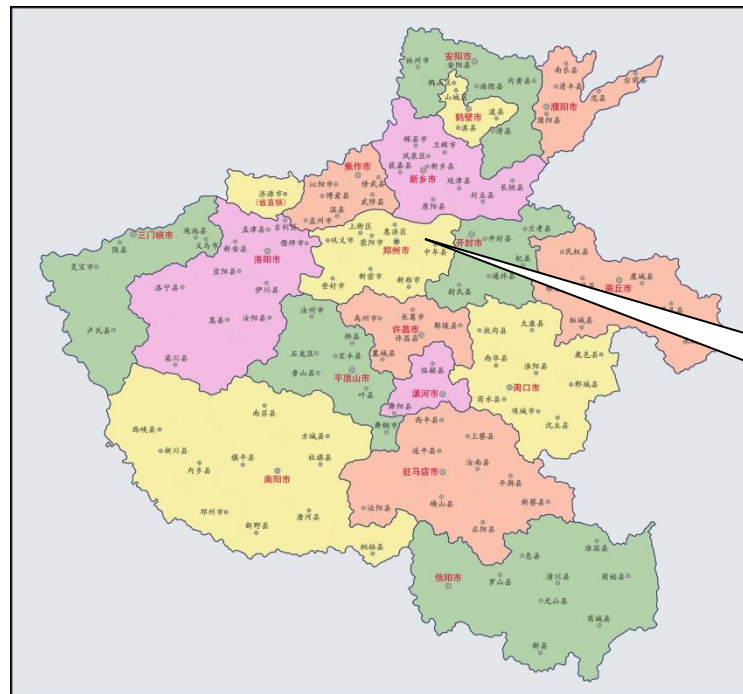
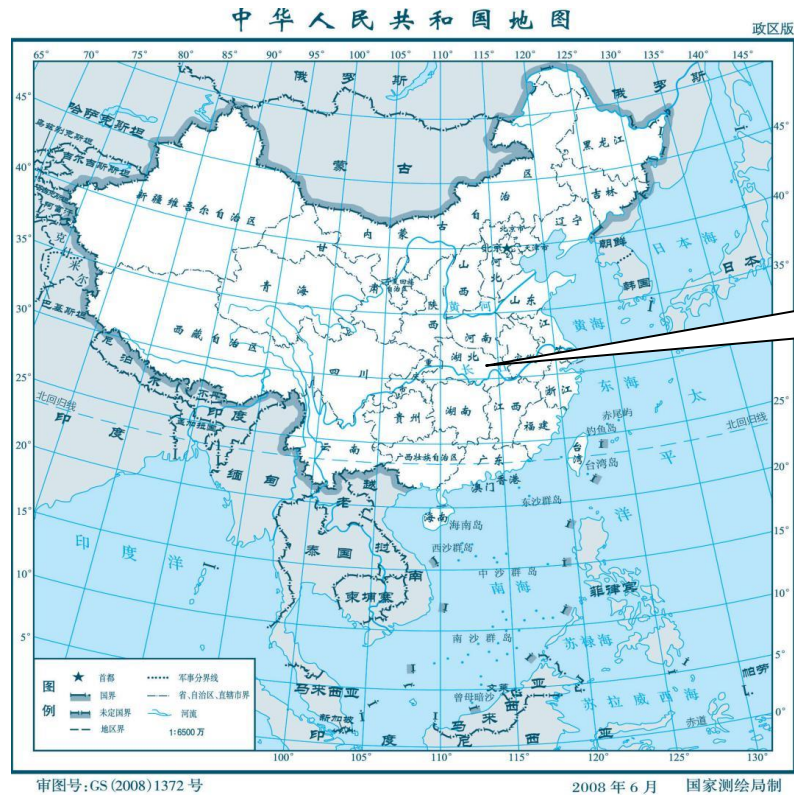




Figure 1.1 Location of the project

1.9 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.H	AMS-III.H. Methane recovery in wastewater treatment	19.0
Methodology	AMS-I.D	AMS-I.D. Grid connected renewable electricity generation	18.0
Tool	07	Tool to calculate the emission factor for an electricity system	07.0
Tool	01	Tool for the demonstration and assessment of additionality	07.0.0
Tool	21	Demonstration of additionality of small-scale project activities	13.1
Tool	05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	3.0

1.10 Double Counting and Participation under Other GHG Programs

1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.10.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.12 Sustainable Development Contributions

The project is located in Zhengyoumo Village, Zhengnan Town, Zhongmu County, Zhengzhou City, Henan Province, P.R. China. The project captures leachate biogas and generates electricity, which is fed into the CCPG, displacing energy that would otherwise be generated by fossil fuels.

The Project contributes to sustainable development in the following ways:

- SDG 13 Climate Action

The project will recover and destroy leachate biogas that consists mainly of greenhouse gas methane and would otherwise be released directly into the atmosphere, effectively reducing greenhouse gas emissions, which contributes to the China's commitment to peak carbon dioxide emissions before 2030. Total GHG emission reductions and removals of 663,100 tCO₂e will be achieved during the fixed 10 years crediting period.

- SDG 8 Decent works and Economic Growth

The project can provide permanent and full-time job opportunities over the project lifetime.

- SDG 7 Affordable and Clean Energy

The project utilizes the leachate biogas from Zhengzhou (East) environmental Protection Energy Project to generate electricity by gas engines. Total electricity generation supplied by the project to the grid is estimated to be 118,470 MWh, which could replace the equivalent electricity from fossil fuel based grid.

Table 1: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	The project avoided anthropogenic emissions of greenhouse gases (GHG) of 123,622 tCO ₂ e during the monitoring period (21-Jan-2022 to 31-Dec-2023)	Prevented the release of 123,622 tonnes of carbon into the atmosphere over project lifetime (21-Jan-2022 to 31-Dec-2023)
2)	8.5	Total number of job opportunities	Implemented activities to increase	The project activity employs 10 permanent and full-time employees through its operational company Zhongmu Branch company of Henan BCCY Environmental Energy Co., Ltd. during the monitoring period (21-Jan-2022 to 31-Dec-2023)	Increased 10 permanent and full-time job opportunities over project lifetime (21-Jan-2022 to 31-Dec-2023)
3)	7.2	MWh of renewable energy generated	Implemented activities to increase	Total 20,740.137 MWh of renewable energy generated during the monitoring period (21-Jan-2022 to 31-Dec-2023)	Generated 20,740.137 MWh of renewable energy over project lifetime (21-Jan-2022 to 31-Dec-2023)

1.13 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

Since the project is located in Zhengang Town, Zhongmu County, Zhengzhou City, Henan Province, P. R. China, local residents in Zhengang Town are considered as stakeholders. Since the project is to collect the leachate biogas from Zhengzhou (East) environmental Protection Energy Project and utilize it to generate electricity, employees of Zhengzhou (East) environmental Protection Energy Project are considered as stakeholders.

The stakeholder information is described below:

The stakeholder information is described below:				
Item			Quantity	Percentage
1	Gender	Male	25	83%
		Female	5	17%
2	Age	<25	2	6%
		25-45	23	77%
		45-55	4	14%
		>55	1	3%
3	Education	Junior high school or below	8	27%
		Senior high school	12	40%
		College or above	10	33%
4	Occupation	Worker	16	53%
		Peasant	8	27%
		Management Personnel	2	7%
		Individual household	4	13%

Legal or customary tenure/access rights

The project area is in the Zhengang Town, Zhongmu County, Zhengzhou City, Henan Province, P.R. China and does not involve the land issues.

	According to the cooperation agreement signed between the project owner and Zhengzhou Dongxing Environmental Protection Energy Co., Ltd, the project owner has the right to use their leachate biogas for power generation. There are no uncertainties related to legal or customary tenure/access rights to territories and resources, including collective and conflicting rights.
Stakeholder diversity and changes over time	Stakeholders identified for this project activity are diverse in nature – in terms of socio-economic status, age and gender wise and the diversity is unlikely to change over time.
Expected changes in well-being	The project is expected to improve local environment, provide work opportunities, improve the operational safety of the waste incineration plant and supply clean energy.
Location of stakeholders	The location of stakeholder is in the Zhongmu County, Zhengzhou City, Henan Province, P.R. China. There is no areas outside the project area that is predicted to be impacted by the project.
Location of resources	The location is in the Zhongmu County, Zhengzhou City, Henan Province, P.R. China.

2.1.2 Stakeholder Consultation and Ongoing Communication

Ongoing consultation	Before the implementation of project, the stakeholders were informed about the stakeholder meeting through posters on 10-Nov-2021. In the bulletin, the company invited the potential stakeholders to get to know their opinions and/or suggestions about the implementation of the project and VCS application of the project. The stakeholder consultancy meeting for the parties interested in the project was organized at Dongxing Waste Incineration Plant on 25-Nov-2021 to collect opinions from all stakeholders, such as representatives of local residents, Dongxing Waste Incineration Plant, the project owner and so on. The project owner appointed one person to make a meeting minute. During the stakeholder consultancy meeting, some stakeholders worried that the project will bring noise, land occupying and pollution on employees and local residents' living. For these issues, the project owner will offer some
----------------------	---

	measures: i.e. installation of sound proof devices and plating of green isolation belts are used for mitigating noise; the area the project built on only will not occupy farms of local residents; at the project site, waste recycling will be carried out reduce emit of waste, and the project site meanwhile will be far from the residential area, so the effect of the noise from the plant is little to local residents. All attendances to the meeting were satisfied to the measures. During the monitoring period permanent communication has been maintained with the community. The mechanism for on-going communication with local stakeholders include the following channel: Periodic meetings we representatives of Zhongmu Branch company of Henan BCCY Environmental Energy Co., Ltd. The meetings were organized at Dongxing Waste Incineration Plant to collect comments from all stakeholders, such as representatives of local residents, Dongxing Waste Incineration Plant management station, and the project owner and so on. The project owner appointed one person to make a meeting minute. Project proponent checked the comments in the book on a regular basis, and record responses. The project information with contact information was also posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information.
Date(s) of stakeholder consultation	During this monitoring period, 2 meetings were organized at Dongxing Waste Incineration Plant on 16-Aug-2022 and 22-Jun-2023 to collect comments.
Communication of monitored results	As a whole, the impacts of operation of the project are basically positive. All stakeholders were pleased with the development of the project. The VCS project would actually facilitate the development of the local economy and increase the income of local residents. In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals. During this monitoring period, there were no negative comments received for the project.

Consultation records	At the stakeholder meeting on 16-Aug-2022. This meeting was introduced by Tao Jinlong, the plant manager, who described the scale of the project construction, the process flow, the operation of the project and the impact of the application as an emission reduction project on the company. The director of the landfill plant thought that the project had significantly improved the air quality of the landfill during its operation and reduced safety hazards, and that the wastewater and solid waste generated by the project's operation were handled in strict accordance with the requirements to achieve the expected results. The villagers raised questions about whether the project would generate noise in the future, and the plant manager said that the relevant measures would be further strengthened in the future to optimize the reduction of noise. The villagers unanimously praised the project at this meeting. At the stakeholder meeting on 22-Jun-2023. This meeting consisted of a main presentation by the plant manager about the project during its operation. The landfill employees expressed their views on the project, he believed that the project could be implemented in accordance with the relevant regulations, and through grid-connected power generation, it could achieve energy saving and emission reduction, which is in line with the national dual-carbon target. The villagers asked questions about whether the project could provide more job opportunities in the future and the environmental impacts generated, and the plant manager said that the demand for jobs in the future would be prioritized to the nearby villagers, that the environment of the plant would be implemented in strict accordance with the requirements, and that the equipment management would be put in place, and that it would not have an impact on the lives of the residents.
Stakeholder input	During the consultation, all stakeholders were supportive of the project, there were no negative comments received for the project. There are no updates to the project design.

2.1.3 Free, Prior, and Informed Consent

Consent	Based on the principle of friendly consultation and on the basis of truly and fully expressing their respective wishes, the project owner and Zhengzhou Dongxing Environmental Protection Energy Co., Ltd signed a cooperation agreement on this project. Based on questionnaire surveys and stakeholder consultation meetings, local residents have expressed support for the project. The enterprise investment project record certificate was approved by the Management Bureau of Zhongmou automobile industry cluster area Management Committee on 16-Sep-2021. An environmental impact assessment (EIA) of the project was approved by Ecological Environment Bureau of Zhengzhou County on 15-Oct-2021. There are no ongoing or unresolved conflicts and the project does not exacerbate nor influence the outcomes of unresolved conflicts.
Outcome of FPIC	The enterprise investment project record certificate was approved by the Management Bureau of Zhongmou automobile industry cluster area Management Committee on 16-Sep-2021. An environmental impact assessment (EIA) of the project was approved by Ecological Environment Bureau of Zhengzhou County on 15-Oct2021. The project is located at Zhengan Town, Zhongmu County, Zhengzhou City, Henan Province, P.R. China. According to the project cooperation agreement, the project has the right to use the land, and there is no violation of the rights of other third parties. Based on questionnaire surveys and stakeholder consultation meetings, local residents have expressed support for the project. The project was constructed with the consent of government departments at all levels. Local stakeholders are supportive

	of construction of the project. A majority of surveyed stakeholders think the project will improve local economic development without adverse environmental impact. The survey shows that the stakeholders are supportive to the project. The project will not encroach on land, relocate people without consent, and force physical or economic displacement.
--	--

2.1.4 Grievance Redress Procedure

Grievances received	Resolution and outcome
No negative grievance was received during the monitoring period.	Not applicable
During the stakeholder meeting, we have told the stakeholders about the grievance procedure of the project. They all agreed with the grievance procedure and felt the procedure was easily accessible to them for ongoing consultation.	

2.1.5 Public Comments

There are no updates to the project design.

Summary of comments received	Actions taken
No negative comments were received during the stakeholder consultation or outside of the public comment period.	Not applicable

2.2 Risks to Stakeholders and the Environment

	Risk identified	Mitigation or preventative measure taken
Risks to stakeholder participation	No risk identified	Based on the stakeholder meeting, the stakeholders involved in the project are very supportive of the project. The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. Therefore, the project has no risks to stakeholder participation.

Working conditions	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. The Project's purpose is to supply electricity from the methane recovery of biogas to replace fossil fuel power. It is beneficial to local people. The Project has DCS system. Therefore, most of the employees work in indoor environment (at the office), instead of having to stand outside. In case of on-site monitoring and device maintenance, the project owner provides safety protection equipment.
Safety of women and girls	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. Workers including women will be employed during construction and operation period and the project has no impact to safety of women and girls. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specifically regarding the gender equality, detailed enforcement rules are regulated in "Women's Rights Protection Law²" to provide support and benefits to women.
Safety of minority and marginalized groups, including children	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. China has published protection laws on the minors³. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the biogas plant, as opposed to manufacturing industries, which may require abundant low-skilled labor. Therefore, the project does not employ and is not complicit in any form of child labor. According to the Employee list, it could be confirmed that the project has no impact to safety of minority and marginalized groups, including children.

² https://www.gov.cn/guoqing/2021-10/29/content_5647634.htm

³ http://www.npc.gov.cn/zgrdw/npc/zfjc/zfjcelys/2014-05/08/content_1862530.htm

Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	No risk identified	An environmental impact assessment (EIA) of the project was approved by Ecological Environment Bureau of Zhengzhou County on 15-Oct-2021. Measures has been identified in EIA of the project to mitigate environment pollution caused by construction and operation of the project, which will be implemented as per environment pollution related laws and regulations and supervised by local government. During construction phase, the raise dust, noise, waste water and solid wastes caused by project construction have been treated according to the measures in EIA report, and there is no significant impact on the environment. During the operation period, all the mitigation measures proposed by EIA report have been implemented and the following key aspects are addressed: Waste water The wastewater produced by the project consists of biogas condensation waste water, circulating cooling waste water of generator set, and reclaimed waste water of soft water equipment. Condensation wastewater was treated by leachate treatment system of Zhengzhou Dongxing Environmental Protection Energy Co., LTD., the sewage is reused and not discharged. Therefore, it can meet the standards of "The reuse of urban recycling water- Waste quality standard for industrial uses"(GB/T19923-2005). Waste Gas The exhaust gas produced by this project is mainly the exhaust gas of the generators. The concentration of the exhaust gas (mainly SO₂, NO_x and particulate matter) after combustion in gas-generator sets and processed by the SCR denitration system is discharged through the 15m high cylinder exhaust. The exhaust gas meets the secondary standards in Table 2 of the Comprehensive Emission Standard for Air Pollutants (GB16297-1996) (particulate matter
---	--------------------	---

	$\leq 10\text{mg}/\text{m}^3$, $\text{SO}_2 \leq 35\text{mg}/\text{m}^3$, $\text{NO}_x \leq 50\text{mg}/\text{m}^3$). All relevant data are listed in the test report issued by Henan Provincial Government Testing Co., LTD. Noise pollution According to the prediction results, the acoustic environment quality of the project equipment noise transmitted to the northwest boundary of project can meet the Class 2 Zone standards of "Environmental Noise Emission Standard of Industrial Enterprise Boundary" (GB12348-2008). The acoustic environment quality of the project equipment noise transmitted to the southeast boundary of project can meet the Class 4 Zone standards of "Environmental Noise Emission Standard of Industrial Enterprise Boundary" (GB12348-2008). All relevant data are listed in the test report issued by Henan Provincial Government Testing Co., LTD. Solid waste The solid waste of the project is mainly sulfur paste, waste oil, waste oil barrel and waste denitration catalyst. The sulfur paste is stored in the sulfur storage room and sold regularly. The waste oil, waste oil barrels and waste denitration catalyst are stored in the hazardous waste room of the factory, and regularly handed over to qualified units for recycling and processing. The construction standard of hazardous waste room shall meet the requirements of "Standard for pollution control on hazardous waste storage" (GB18597-2001). The project signed a hazardous waste collection and disposal contract with Henan Junchi Environmental Protection Technology Co., LTD. Specific hazardous waste disposal information is listed in the hazardous waste disposal joint sheet. According to the environmental monitoring report and the hazardous waste disposal joint order, the annual discharge of major pollutants of the project has reached the standard, and the pollution facilities of the project have been
--	---

improved and the pollution control measures are effective.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Discrimination and sexual harassment	No discrimination or sexual harassment occur. China has published “Labor Law” ⁴ and “Women’s Rights Protection Law”, which could prohibit such phenomena. The project abides the rules of equality and does not involve and is not complicit in any form of discrimination.
Management experience	Parent company of the PP operates more than 100 similar projects ⁵ , so it has sufficient management and operational experience. Where relevant experience is lacking, the PP has a sound recruitment system to fill the identified gaps. Before the construction of the project, the project owner signed a contract with a professionally qualified construction unit.
Gender equity in labor and work	As a project combined of positive environmental, economic, and sustainable development benefits, the project is to generate clean electricity from the biogas sources to replace fossil fuel power. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in “Women’s Rights Protection Law” to provide support and benefits to women. Qualified local residents, both men and women, are recruited to work for the project. The project follows the provisions of the “Constitution”⁶ and “Labor Law” on equal pay for men and women for equal work. During stakeholders’ consultation process, comments were collected from the local people, including both men and women. The project abides the rules of equality. Regarding the gender equality, detailed enforcement rules are regulated in “Labor Law” and “Women’s Rights Protection Law”.

⁴ https://www.gov.cn/banshi/2005-05/25/content_905.htm

⁵ <https://www.bccynewpower.com/about/0>

⁶ https://www.gov.cn/guoqing/2018-03/22/content_5276318.htm

Human trafficking, forced labor, and child labor

The project does not use victims of human trafficking, forced labor, and child labor. Human trafficking is a serious crime that is strictly regulated and combated in the Chinese legal system. The relevant laws and regulations mainly include the “Criminal Law”⁷ and the “Criminal Procedure Law”⁸.

China has promulgated the “Labor Law” to protect the legitimate rights and interests of workers and prohibit forced labor.

China has published protection laws on the minors. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the power plant. Therefore, the project does not employ and is not complicit in any form of child labour. According to the Employee list, it could be confirmed that the Project does not employ any child labour.

2.3.2 Human Rights

The project respects internationally proclaimed human rights. China has its own legislation in place prohibiting the violation of human rights principle and it actively enforces the compliance of such principle. China has ratified two UN human rights treaties:

1. The International Covenant on Civil and Political Rights, the International Covenant on Economic, Social, and Cultural Rights. China has published relevant regulations and practice into line with the treaties, such as Labour Contract Law⁹.
2. This project does not discriminate with regards to participation and inclusion. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specifically regarding the gender equality, detailed enforcement rules are regulated in ‘Women’s Rights Protection Law’¹⁰ to provide support and benefits to women.

2.3.3 Indigenous Peoples and Cultural Heritage

According to the EIA reports, the Project area does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture. There are no indigenous peoples present in or within the area of influence of the Project.

2.3.4 Property Rights

⁷ <https://flk.npc.gov.cn/detail2.html?ZmY4MDgxODE3OTZhNjM2YTAxNzk4MjJhMTk2NDBjOTI>

⁸ <https://flk.npc.gov.cn/detail2.html?ZmY4MDgwODE2ZjEzNWY0NjAxNmYxZDFiODFiMDEzNTE%3D>

⁹ https://www.gov.cn/flfg/2007-06/29/content_669394.htm

¹⁰ https://www.gov.cn/banshi/2005-05/26/content_980.htm

Disputes over rights to territories and resources	N/A
Respect for property rights	The project has obtained necessary approval from the local government. The project has no disputes occurred, therefore measures implemented to protect and preserve the property rights of stakeholders, IPs, LCs, and customary rights holders are not applicable.

2.3.5 Benefit Sharing

Summary of the benefit sharing plan	As the project does not impact property rights as described in Section 2.3.4 above, the benefit sharing plan information is not applicable.
Benefit sharing during the monitoring period	As the project does not impact property rights as described in Section 2.3.4 above, the benefit sharing information is not applicable.

2.4 Ecosystem Health

	Risk identified	Mitigation or preventative measure taken during the monitoring period
Impacts on biodiversity and ecosystems	No risk identified	The project is biogas recovery and utilization project. According to the EIA report, it does not affect biodiversity and ecosystems.
Soil degradation and soil erosion	No risk identified	The Project's purpose is to supply electricity from biogas. Therefore, the impact of the Project on soil degradation and soil erosion is negligible.
Water consumption and stress	No risk identified	The Project's purpose is to supply electricity from biogas. Therefore, the impact of the Project on water consumption and stress is negligible.
Usage of fertilizers	No risk identified	The project is a biogas recovery and utilization project. Therefore, the Project does not involve the fertilizer utilization.

2.4.1 Rare, Threatened, and Endangered species

According to EIA, the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species. Therefore, the following table isn't applicable.

Species or habitat	Not applicable
--------------------	----------------

2.4.2 Introduction of species

According to EIA, no planting or species introduction are in the project. This section is left as N/A.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A	N/A	N/A	N/A

According to EIA, no invasive species exist in the project area. Therefore, the following table isn't applicable.

Existing invasive species	Mitigation measures to prevent spread or continued existence of invasive species
N/A	N/A

2.4.3 Ecosystem conversion

Not applicable as the project is not a ARR, ALM, WRC or ACoGS project.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The project started to construct on 15-Sep-2021 and started operation on 21-Jan-2022. During the monitoring period, the project had been running well. The total installed capacity is 3MW, including 3 sets with each capacity of 1MW.

Table 3.1 Relevant dates for the project activity

No.	Project activity	Date
-----	------------------	------

1	Operation of engine 1#	26/Apr/2022
2	Operation of engine 2#	21/Jan/2022
3	Operation of engine 3#	28/Apr/2022

The project consists in leachate biogas collection, transmission and treatment system, with subsequent electricity generation and grid connection system.

Leachate biogas collection system

Leachate biogas is extracted from UASB anaerobic biochemical system by the top set gas pipe collection into the storage tank. The pipeline is connected to the biogas storage tank through Roots fan to extract and transport biogas into the biogas pretreatment unit. Flow rate of the leachate biogas is regulated at the collection points in order to always fit with the consumption capacity of the generation engines.

Leachate biogas treatment system

Prior to electricity generation, leachate biogas is treated to remove impurities and moisture, to avoid corrosion in the engines. The treatment consists of filtration, de-moisturing, cooling and pressurization.

Electricity generation and grid connection system

3 gas engines of 1MW rated power each are fed with the leachate biogas and generate electricity, which is then exported to the grid.

Table 3.2 Equipment technical parameters

Main Equipment	Parameter	Value
Gas engine	Type	J320GS
	Manufacturer	INNIO Jenbacher GmbH & Co OG
	Rated capacity	1000 kW
	Rated rotation speed	1500 r/min
	Lifetime	20 years

3.2 Deviations

3.2.1 Methodology Deviations

No methodology deviation is applied in the project.

3.2.2 Project Description Deviations

No project description deviation is applied in the project.

3.3 Grouped Projects

The project is not a grouped project.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes ☒ No

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	$MCF_{ww,treatment,BL,i}$
Data unit	-
Description	Methane correction factor for baseline wastewater treatment systems i
Source of data	MCF values as per Table 2 of AMS-III.H. V.19
Value applied	0.8
Justification of choice of data or description of measurement methods and procedures applied	Default values from chapter 6 of volume 5. Waste in 2006 IPCC Guidelines for National Greenhouse Gas Inventories.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$\eta_{COD,BL,i}$
Data unit	%
Description	COD removal efficiency of the baseline treatment system
Source of data	Operation report
Value applied	76.33%

Justification of choice of data or description of measurement methods and procedures applied	The COD removal efficiency is based on COD values for baseline system before implementation of project activity.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$B_{o,ww}$
Data unit	kg CH ₄ /kg COD
Description	Methane producing capacity of the wastewater
Source of data	Paragraph 29 of AMS-III.H V.19
Value applied	0.25
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value applies in line with Equation 2 of AMS-III.H V.19
Purpose of Data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	UF_{BL}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	Paragraph 29 of AMS-III.H V.19
Value applied	0.89
Justification of choice of data or description of measurement methods and procedures applied	As specified in Paragraph 29 of AMS-III.H V.19
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	UF_{PJ}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.H V.19
Value applied	1.12
Justification of choice of data or description of measurement methods and procedures applied	As specified in methodology AMS-III.H V.19
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	$MCF_{ww,treatment,PJ,k}$
Data unit	-
Description	Methane correction factor for project wastewater treatment system k
Source of data	Table 2 of AMS-III.H V.19
Value applied	0.8
Justification of choice of data or description of measurement methods and procedures applied	As specified in methodology AMS-III.H V.19
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	GWP_{CH_4}
Data unit	tCO ₂ e/t CH ₄
Description	Global warming potential of CH ₄

Source of data	IPCC
Value applied	Default value of 28. Shall be updated according to any future COP/CMP decisions
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	CFE_{ww}
Data unit	-
Description	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems
Source of data	Default value as per paragraph 40 Eq 10 of AMS-III.H V.19
Value applied	0.9
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO ₂ /MWh
Description	Operation margin emission factor of CCPG
Source of data	2019 Bulletin on the Baseline Emission Factors of the China Grids
Value applied	0.8587
Justification of choice of data or description of measurement methods and procedures applied	The value is published by Ministry of Ecology and Environment of the People's Republic of China, based on China Electric Yearbook (2016-2018), Electricity Industry Statistics (2015-2017), China Energy Statistic Yearbook (2016-2018) and IPCC data.

Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO ₂ /MWh
Description	Build margin emission factor of CCPG
Source of data	2019 Bulletin on the Baseline Emission Factors of the China Grids
Value applied	0.2854
Justification of choice of data or description of measurement methods and procedures applied	The value is published by Ministry of Ecology and Environment of the People's Republic of China, based on China Electric Yearbook (2016-2018), Electricity Industry Statistics (2015-2017), China Energy Statistic Yearbook (2016-2018) and IPCC data.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$TDL_{j,y}$
Data unit	%
Description	Average technical transmission and distribution losses for providing electricity to source j,k in year y
Source of data	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)
Value applied	20%
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	D _{CH4}
Data unit	Tonnes /m ³
Description	Methane density
Source of data	Tool 08 “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (version 03.0)
Value applied	0.0007156
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	At the normal conditions (273.15 K and 101,325 Pa)

Data / Parameter	FE
Data unit	%
Description	Flare efficiency in year y
Source of data	Default value as per paragraph 45 Eq 10 of AMS-III.H V.19
Value applied	100
Justification of choice of data or description of measurement methods and procedures applied	If the biogas is combusted for gainful purposes, e.g. fed to an engine, an efficiency of 100 per cent may be applied
Purpose of Data	Calculation of baseline emissions
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	Q _{ww,i,y}
Data unit	m ³ /month

Description	The flow of wastewater																																																																			
Source of data	Continuous flow meter																																																																			
Description of measurement methods and procedures to be applied	Measured continuously by flow meters installed before entering the treatment system. The flow of wastewater is the sum of F1 and F2.																																																																			
Frequency of monitoring/recording	Parameter monitored continuously but aggregated monthly and annually for calculations.																																																																			
Value applied	<table><tr><th>Date</th><th>South zone flow of wastewater Q_{north} (m³)</th><th>North zone flow of wastewater Q_{south} (m³)</th><th>The flow of wastewater $Q_{ww,i,y}$ (m³)</th></tr><tr><td>21-Jan-2022-31-Jan-2022</td><td>5,258.64</td><td>5,082.96</td><td>10,341.60</td></tr><tr><td>01-Feb-2022-28-Feb-2022</td><td>10,887.89</td><td>11,600.16</td><td>22,488.05</td></tr><tr><td>01-Mar-2022-31-Mar-2022</td><td>10,660.45</td><td>10,368.16</td><td>21,028.61</td></tr><tr><td>01-Apr-2022-31-Apr-2022</td><td>13,300.71</td><td>12,521.47</td><td>25,822.18</td></tr><tr><td>01-May-2022-31-May-2022</td><td>10,573.85</td><td>10,943.30</td><td>21,517.15</td></tr><tr><td>01-Jun-2022-31-Jun-2022</td><td>15,303.26</td><td>15,332.08</td><td>30,635.34</td></tr><tr><td>01-Jul-2022-31-Jul-2022</td><td>18,404.81</td><td>18,885.66</td><td>37,290.47</td></tr><tr><td>01-Aug-2022-31-Aug-2022</td><td>17,783.16</td><td>17,806.50</td><td>35,589.66</td></tr><tr><td>01-Sep-2022-31-Sep-2022</td><td>17,010.10</td><td>16,303.25</td><td>33,313.35</td></tr><tr><td>01-Oct-2022-31-Oct-2022</td><td>11,172.08</td><td>11,725.71</td><td>22,897.79</td></tr><tr><td>01-Nov-2022-31-Nov-2022</td><td>10,689.69</td><td>9,403.24</td><td>20,092.93</td></tr><tr><td>01-Dec-2022-31-Dec-2022</td><td>6,742.11</td><td>7,164.04</td><td>13,906.15</td></tr><tr><td>01-Jan-2023-31-Jan-2023</td><td>13,164.71</td><td>12,848.19</td><td>26,012.90</td></tr><tr><td>01-Feb-2023-28-Feb-2023</td><td>9,487.51</td><td>9,755.72</td><td>19,243.23</td></tr><tr><td>01-Mar-2023-31-Mar-2023</td><td>12,251.27</td><td>12,008.26</td><td>24,259.53</td></tr></table>				Date	South zone flow of wastewater Q_{north} (m ³)	North zone flow of wastewater Q_{south} (m ³)	The flow of wastewater $Q_{ww,i,y}$ (m ³)	21-Jan-2022-31-Jan-2022	5,258.64	5,082.96	10,341.60	01-Feb-2022-28-Feb-2022	10,887.89	11,600.16	22,488.05	01-Mar-2022-31-Mar-2022	10,660.45	10,368.16	21,028.61	01-Apr-2022-31-Apr-2022	13,300.71	12,521.47	25,822.18	01-May-2022-31-May-2022	10,573.85	10,943.30	21,517.15	01-Jun-2022-31-Jun-2022	15,303.26	15,332.08	30,635.34	01-Jul-2022-31-Jul-2022	18,404.81	18,885.66	37,290.47	01-Aug-2022-31-Aug-2022	17,783.16	17,806.50	35,589.66	01-Sep-2022-31-Sep-2022	17,010.10	16,303.25	33,313.35	01-Oct-2022-31-Oct-2022	11,172.08	11,725.71	22,897.79	01-Nov-2022-31-Nov-2022	10,689.69	9,403.24	20,092.93	01-Dec-2022-31-Dec-2022	6,742.11	7,164.04	13,906.15	01-Jan-2023-31-Jan-2023	13,164.71	12,848.19	26,012.90	01-Feb-2023-28-Feb-2023	9,487.51	9,755.72	19,243.23	01-Mar-2023-31-Mar-2023	12,251.27	12,008.26	24,259.53
Date	South zone flow of wastewater Q_{north} (m ³)	North zone flow of wastewater Q_{south} (m ³)	The flow of wastewater $Q_{ww,i,y}$ (m ³)																																																																	
21-Jan-2022-31-Jan-2022	5,258.64	5,082.96	10,341.60																																																																	
01-Feb-2022-28-Feb-2022	10,887.89	11,600.16	22,488.05																																																																	
01-Mar-2022-31-Mar-2022	10,660.45	10,368.16	21,028.61																																																																	
01-Apr-2022-31-Apr-2022	13,300.71	12,521.47	25,822.18																																																																	
01-May-2022-31-May-2022	10,573.85	10,943.30	21,517.15																																																																	
01-Jun-2022-31-Jun-2022	15,303.26	15,332.08	30,635.34																																																																	
01-Jul-2022-31-Jul-2022	18,404.81	18,885.66	37,290.47																																																																	
01-Aug-2022-31-Aug-2022	17,783.16	17,806.50	35,589.66																																																																	
01-Sep-2022-31-Sep-2022	17,010.10	16,303.25	33,313.35																																																																	
01-Oct-2022-31-Oct-2022	11,172.08	11,725.71	22,897.79																																																																	
01-Nov-2022-31-Nov-2022	10,689.69	9,403.24	20,092.93																																																																	
01-Dec-2022-31-Dec-2022	6,742.11	7,164.04	13,906.15																																																																	
01-Jan-2023-31-Jan-2023	13,164.71	12,848.19	26,012.90																																																																	
01-Feb-2023-28-Feb-2023	9,487.51	9,755.72	19,243.23																																																																	
01-Mar-2023-31-Mar-2023	12,251.27	12,008.26	24,259.53																																																																	

	01-Apr-2023-31-Apr-2023	13,781.68	12,850.70	26,632.38		
	01-May-2023-31-May-2023	10,357.39	10,259.23	20,616.62		
	01-Jun-2023-31-Jun-2023	14,291.66	14,088.47	28,380.13		
	01-Jul-2023-31-Jul-2023	17,694.01	18,451.67	36,145.68		
	01-Aug-2023-31-Aug-2023	18,067.42	18,103.10	36,170.52		
	01-Sep-2023-31-Sep-2023	16,290.21	15,997.47	32,287.68		
	01-Oct-2023-31-Oct-2023	11,974.77	11,939.11	23,913.88		
	01-Nov-2023-31-Nov-2023	12,367.73	11,578.22	23,945.95		
	01-Dec-2023-31-Dec-2023	8,793.30	9,015.45	17,808.75		
	Total	306,308.41	304,032.12	610,340.53		
Monitoring equipment	Continuous Flow Meter					
	Nam e	Model	Accurac y	Serial Number	Date of calibration	Validit y
	North zone Flow Meter F1	PROMA G 10	0.5	R70DB318000	28-Oct-2021	27-Oct-2022
					27-Oct-2022	26-Oct-2023
					26-Oct-2023	25-Oct-2024
	South zone Flow Meter F2	PROMA G 10	0.5	S9040018000	28-Oct-2021	27-Oct-2022
					27-Oct-2022	26-Oct-2023
					26-Oct-2023	25-Oct-2024

QA/QC procedures to be applied	Zhengzhou Dongxing Environmental Protection Energy Co., Ltd. employs an internal QA audit process that ensures monitoring activities are conducted in accordance with the monitoring plan and verifies the accuracy of data reported. The flow meter will be operated and calibrated according to manufacturer's specifications but no less than every 3 years). A calibration/service log will be maintained for the flow meter.
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	BG _{burnt,y}					
Data unit	Nm ³					
Description	Biogas volume in year y					
Source of data	Daily records of the flow meter F3					
Description of measurement methods and procedures to be applied	Measured continuously by flow meters installed before entering the treatment system.					
Frequency of monitoring/recording	Measured continuously by a flow meter, and recorded at least once per hour. Data to be aggregated monthly and yearly.					
Value applied	11,902,032.00					
	Date		BG _{burnt,y} (m ³)			
	21-Jan-2022-31-Dec-2022		5,136,216.00			
	21-Jan-2023-31-Dec-2023		6,765,816.00			
	Total		11,902,032.00			
Monitoring equipment	Continuous Flow Meter					
	Name	Model	Accuracy	Serial Number	Date of calibration	Validity
	Flow Meter F3	Prosonic Flow B	1.5	S91A7C02000	13-Sep-2021	12-Sep-2022

					12-Sep-2022	11-Sep-2023
					11-Sep-2023	10-Sep-2024
QA/QC procedures to be applied	The calibration is being done once a year.					
Purpose of data	Calculation of baseline and project emissions					
Calculation method	-					
Comments	Flow meters automatically measure temperature and pressure, expressing biogas volumes in Normalized cubic meters (Nm ³). Data are archived during the crediting period and kept until two years after.					

Data / Parameter	WCH ₄ .y					
Data unit	%					
Description	Methane content in biogas in the year y					
Source of data	Daily records of the gas analyzer meter					
Description of measurement methods and procedures to be applied	Measured directly by continuous gas analyzer.					
Frequency of monitoring/recording	Measurements are recorded electronically at least once per day. The data are measured and archived electronically.					
Value applied	65.19%					
Monitoring equipment	Gas analyzer					
	Name	Type	Accuracy class	Serial number	Date of calibration	Validity
	Gas analyzer	BIOGAS 5000	0.5%	G502590	09-Nov-2021	08-Nov - 2022

				08-Nov-2022	07-Nov - 2023
				07-Nov-2023	06-Nov - 2024
QA/QC procedures to be applied	Gas analyzer is calibrated once a year.				
Purpose of data	Calculation of baseline and project emissions				
Calculation method	-				
Comments	-				

Data / Parameter	COD _{ww,untreated,y}			
Data unit	t COD/m ³			
Description	The chemical oxygen demand of the wastewater before the treatment system affected by the project activity			
Source of data	Sampling analysis			
Description of measurement methods and procedures to be applied	The COD content will be analyzed using a spectrophotometric method. The results will be logged in the plant operation report on a daily basis.			
Frequency of monitoring/recording	Measured and recorded every day. The confidence/precision level of 90/10 shall be attained.			
Value applied	0.057818			
	Date	COD _{ww,untreated,south} (mg/L)	COD _{ww,untreated,north} (mg/L)	COD _{ww,untreated,y} (mg/L)
	21-Jan-2022-31-Dec-2022	58,623.78	55,367.88	56,995.83
				0.056996

	21-Jan-2023-31-Dec-2023	58,495.24	58,784.12	58,639.68	0.058640	
	Average				0.057818	
Monitoring equipment	UV756 UV spectrophotometer					
	Name	Type	Serial number	Accuracy class	Date of calibration	Validity
	Spectrophotometer	UV756	YD03041905012	0.5nm	28-Oct-2021	27-Oct-2022
					27-Oct-2022	26-Oct-2023
					26-Oct-2023	25-Oct-2024
QA/QC procedures to be applied	Measure the COD in accordance with Technical requirement and test procedures for rapid COD photometric analyzer (HJ924-2017).					
Purpose of data	Calculation of baseline and project emissions					
Calculation method	-					
Comments	-					

Data / Parameter	COD _{ww,treated,y}
Data unit	t COD/m ³
Description	The chemical oxygen demand of the wastewater after the treatment system affected by the project activity
Source of data	Sampling analysis
Description of measurement methods and procedures to be applied	The COD content will be analyzed using a spectrophotometric method. The results will be logged in the plant operation report on a daily basis.

Frequency of monitoring/recording	Measured and recorded every day. The confidence/precision level of 90/10 shall be attained.					
Value applied	0.014907					
	Date	COD _{ww,treated,south} (mg/L)	COD _{ww,treated,north} (mg/L)	COD _{ww,treated,y} (mg/L)	COD _{ww,treated,y} (t/m³)	
	21-Jan-2022-31-Dec-2022	14,889.87	14,924.94	14,907.41	0.014907	
	21-Jan-2023-31-Dec-2023	14,922.03	14,890.42	14,906.81	0.014906	
	Average				0.014907	
Monitoring equipment	UV756 UV spectrophotometer					
	Name	Type	Serial number	Accuracy class	Date of calibration	Validity
	Spectrophotometer	UV756	YD03041905012	0.5nm	28-Oct-2021	27-Oct-2022
					27-Oct-2022	26-Oct-2023
					26-Oct-2023	25-Oct-2024
QA/QC procedures to be applied	Measure the COD in accordance with Technical requirement and test procedures for rapid COD photometric analyzer (HJ924-2017).					
Purpose of data	Calculation of baseline and project emissions					
Calculation method	-					
Comments	-					

Data / Parameter	EGPJ,facility,y				
Data unit	MWh				
Description	Quantity of net electricity generated supplied by the project plant/unit in year y				
Source of data	Electricity meter E				
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter (bi-directional) installed at the project site. All data will be monitored and archived electronically. Double check by receipt of electricity sales.				
Frequency of monitoring/recording	The recording frequency will be hourly measured and record and monthly aggregated.				
Value applied	20,740.137				
	Year		Quantity of net electricity generated supplied by the project plant (MWh)		
	21-Jan-2022-31-Dec-2022		8,950.628		
	21-Jan-2023-31-Dec-2023		11,789.509		
	Total		20,740.137		
Monitoring equipment	Electricity meter E				
	Model	Serial number	Accuracy class	Date of calibration	Validity
	DTS858	RM221100098258	0.2S	31-Aug-2021	30-Aug-2022
				30-Aug-2022	29-Aug-2023
				29-Aug-2023	28-Aug-2024
QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.				
Purpose of data	Calculation of baseline emissions				

Calculation method	-
Comments	-

Data / Parameter	EC _{PJ,y}				
Data unit	MWh				
Description	Amount of electricity consumed by the project activity in year y				
Source of data	Electricity meter E				
Description of measurement methods and procedures to be applied	Sources of consumption include electricity consumed for the operation of the biogas capture system, for any processing and upgrading of the biogas, for transportation of the biogas to the flare or power generators, etc.				
Frequency of monitoring/recording	The recording frequency will be hourly measured and record and monthly aggregated.				
Value applied	2.388				
	Year		Amount of electricity consumed by the project activity (MWh)		
	21-Jan-2022-31-Dec-2022		1.506		
	21-Jan-2023-31-Dec-2023		0.882		
	Total		2.388		
Monitoring equipment	Electricity meter E				
	Model	Serial number	Accuracy class	Date of calibration	Validity
	DTS858	RM221100098258	0.2S	31-Aug-2021	30-Aug-2022
				30-Aug-2022	29-Aug-2023
				29-Aug-2023	28-Aug-2024

QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.
Purpose of data	Calculation of project and baseline emissions
Calculation method	-
Comments	-

Data / Parameter	P
Data unit	Pa
Description	Pressure of the biogas
Source of data	-
Description of measurement methods and procedures to be applied	The pressure of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas.
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	T
Data unit	°C
Description	Temperature of the biogas
Source of data	-

Description of measurement methods and procedures to be applied	. The temperature of the gas is required to determine the density of the methane combusted. If the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalised flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

4.3 Monitoring Plan

1. The requirement of monitoring plan

The Project proponents will monitor the emission reductions by methods, indicators, and frequency required by Monitoring Methodology AMS-III.H. (Version 19.0) and AMS-I.D (Version 18.0) to ensure project ERs are measurable and real. The monitoring methodology is based on direct measurement of the amount of leachate biogas captured and destroyed by the project and electricity generating units.

2. Responsibilities of operational and management structure

The Project proponent will implement this monitoring plan. The plan could be revised according to suggestions from VVB and the practical circumstances, in order to keep it consistent, transparent and conservative during the monitoring process.

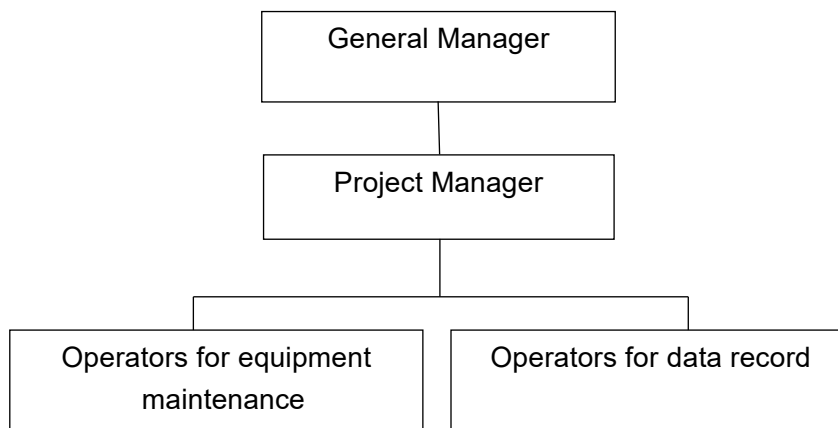


Figure 5.1 Operational and management structure

(1) Principal of the monitoring procedure

The general manager of the project is the leader of the monitoring tasks: he sets out the responsibility of everyone in the monitoring system, and establishes the related documents. The general manager ensures that staff in the monitoring system has the ability to deal with the assigned tasks.

(2) Executive person of the monitoring procedure

Project Manager: a Project Manager is appointed, specifically responsible for training, checking the daily operation, reporting forms and archiving emergency situation reports. The VCS Manager reports monthly to the General Manager (GM) about the project performance and monitored data. In the event that non-conformance in the performance to the mentioned procedures and/or functioning problems of the monitoring equipment are identified, the VCS manager will inform the GM about the situation and work out relevant measures to be taken. The VCS manager will also be responsible for aggregating the monitored data monthly and yearly, archiving and keeping data during the crediting period and two years after.

(3) Operators of the monitoring procedure

Operators will take turns to work in the control center 24 hours a day. They will be in charge of data supervision, filling operation report forms and, checking and inspecting the system. If necessary, they will have the responsibility for executing the emergency plan and drafting emergency situation reports.

3. Monitoring system

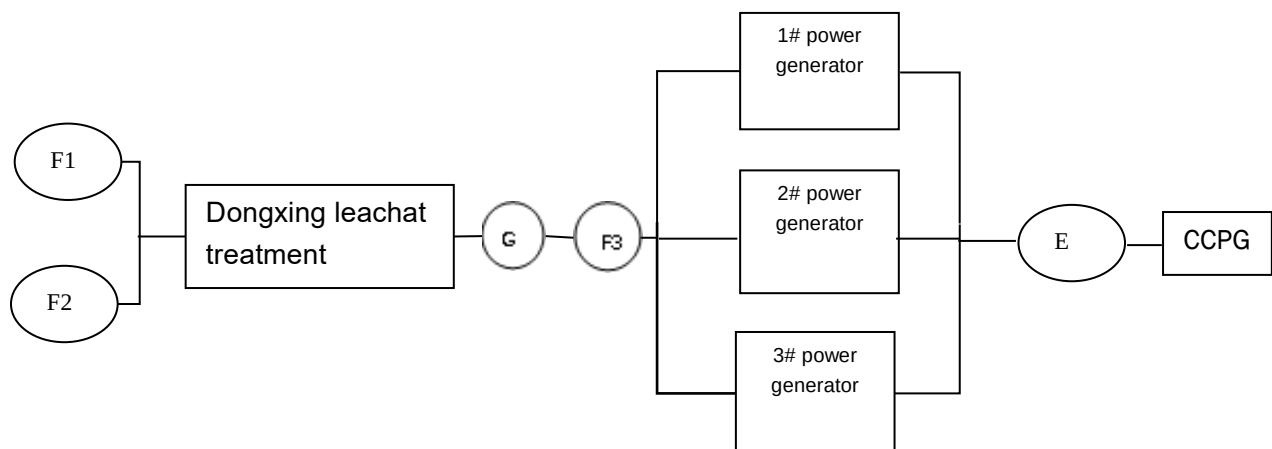


Figure 5.2 Monitoring system

Main parameters:

E	Electricity meter (main meter, bi-directional) to continuously measure the electricity supplied to the CCPG and the electricity supplied by the grid
F1	Flow meter to continuously measure the flow of north zone wastewater before entering the treatment system.
F2	Flow meter to continuously measure the flow of south zone wastewater before entering the treatment system.
F3	Flow meter to continuously measure flow rate of total biogas, with automatic measure adjustment for temperature and pressure.
G	Gas analyzer to continuously measure methane fraction in biogas (W_{CH_4}).

Net electricity exported to the grid:

Electricity meter E is the main meter for monitoring the net electricity supplied to and imported from the grid. Therefore, the net electricity supplied to the grid is calculated as: $EG_{OUT} - EG_{IN}$.

The meter E is installed at the project site and sends data to the DCS. The data was checked daily and totaled monthly. Electricity receipts and invoices was used as cross check during verification. E was calibrated by a qualified third party according to the industry standards and applicable regulations.

4. QA/QC

In order to ensure monitoring plan with high quality, QA/QC measures are carried out in monitoring data recording and checking, equipment calibrating and staff training.

Data recording: all data collected are recorded in electronic files which are regularly backed up. The data are checked by the specific staff every day and reported to General Manager by Project manager monthly.

Equipment calibration and maintenance: Project equipment are subject to regular maintenance and testing according to technical specifications from the manufactures to ensure accuracy and good performance.

If there is a problem with the project's monitoring equipment, the relevant data will be discarded during this period.

5. Emergency procedures

Once a meter in fault, it shall be immediately repaired or replaced with another calibrated meter by a professional engineer, and the biogas or electricity generated during the period of erroneous measurement and replacement of the fault meter shall not be accounted for conservative consideration.

Table 4.1: Detailed calibration information

Name	Model	Serial number	Accuracy class	Date of calibration	Validity
North zone Flow Meter F1	PROMAG 10	R70DB318000	0.5	28-Oct-2021	27-Oct-2022
				27-Oct-2022	26-Oct-2023
				26-Oct-2023	25-Oct-2024
South zone Flow Meter F2	PROMAG 10	S9040018000	0.5	28-Oct-2021	27-Oct-2022
				27-Oct-2022	26-Oct-2023
				26-Oct-2023	25-Oct-2024
Flow Meter F3	Prosonic Flow B	S91A7C02000	1.5	13-Sep-2021	12-Sep-2022
				12-Sep-2022	11-Sep-2023
				11-Sep-2023	10-Sep-2024
Gas analyzer	BIOGAS 5000	G502590	0.5%	09-Nov-2021	08-Nov-2022
				08-Nov-2022	07-Nov-2023
				07-Nov-2023	06-Nov-2024
Spectrophotometer	UV756	YD03041905012	0.5nm	28-Oct-2021	27-Oct-2022
				27-Oct-2022	26-Oct-2023
				26-Oct-2023	25-Oct-2024
Electricity meter E	DTS858	RM221100098258	0.2S	31-Aug-2021	30-Aug-2022
				30-Aug-2022	29-Aug-2023
				29-Aug-2023	28-Aug-2024

6. Procedures used for handling any internal auditing performed and any non-conformities identified:

(1) If the internal audit finds non-conformity, the internal auditor shall issue an "Internal audit nonconformity report" and submit it to the internal audit team leader and management representative.

(2) The internal audit team leader will hand over the Internal Audit Nonconformity Report to the audited department with nonconformity items, and require the rectification measures to be completed within the specified time.

(3) The audited department with nonconforming items shall carry out targeted rectification according to the contents of nonconforming items proposed by the internal auditor.

(4) The internal auditor shall track and confirm the rectification results according to the rectification plan and rectification measures proposed in the Internal Audit Nonconformity Report of the audited department.

No non-conformities were identified during this monitoring period,

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

$$BE_y = BE_{y,1} + BE_{y,2}$$

Equation (1)

Parameter	Description	Unit	21-Jan-2022-31-Dec-2022	2023	Total
BE_y	Baseline emissions	tCO ₂ e	66,334.00	75,253.00	141,587.00
$BE_{y,1}$	Baseline emissions associated with wastewater treatment	tCO ₂ e	61,214.02	68,509.16	129,723.18
$BE_{y,2}$	Baseline emissions associated with electricity generation	tCO ₂ e	5,120.21	6,744.19	11,864.40

$$BE_{y,1} = \sum_i (Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}) \times B_{o,ww} \times UF_{BL} \times GWP_{CH_4}$$

Equation (2)

Parameter	Description	Unit	21-Jan-2022-31-Dec-2022	2023	Total
$BE_{y,1}$	Baseline emissions associated with wastewater treatment	tCO ₂ e	61,214.02	68,509.16	129,723.18
$Q_{ww,i,y}$	Volume of wastewater treated in baseline wastewater treatment system	m ³	294,923.28	315,417.25	610,340.53
$COD_{inflow,i,y}$	Chemical oxygen demand of the wastewater inflow to the baseline treatment system	t/m ³	0.056996	0.058640	/

$\eta_{\text{COD,BL},i}^{11}$	COD removal efficiency of the baseline treatment system	%	73.84%	74.58%	/
$\text{MCF}_{\text{ww,treatment,BL},i}$	Methane correction factor for baseline wastewater treatment systems	/	0.8	0.8	/
$B_{o,\text{ww}}$	Methane producing capacity of the wastewater	kg CH ₄ /kg COD	0.25	0.25	/
UF_{BL}	Model correction factor to account for model uncertainties	/	0.89	0.89	/
GWP_{CH_4}	Global Warming Potential for methane	tCO ₂ e/tCH ₄	28	28	/

$$BE_{y,2} = EG_{PJ,y} \times EF_{\text{grid},y}$$

Equation (3)

Parameter	Description	Unit	21-Jan-2022-31-Dec-2022	2023	Total
$BE_{y,2}$	Baseline emissions associated with electricity generation	tCO ₂ e	5,120.21	6,744.19	11,864.40
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity	MWh	8,950.628	11,789.509	20,740.137
$EF_{\text{grid},y}$	Combined margin CO ₂ emission factor for grid connected power generation	t CO ₂ /MWh	0.57205	0.57205	/

¹¹ According to the paragraph 36 of AMS-III.H v.19.0: In determining baseline emissions using equation (1), historical records of at least one year prior to the project implementation shall be used. This shall include for example the COD removal efficiency of the wastewater treatment systems, the amount of dry matter in sludge, power and electricity consumption per m³ of wastewater treated the amount of final sludge generated per tonne of COD removed, and all other parameters required for determination of baseline emissions. Therefore, the data of 2021 is used for the ex-ante calculated of this parameter in PD. The monitoring period is from 21-Jan-2022 to 31-Dec-2023, and the measured values are used for this parameter.

5.2 Project Emissions

$$PE_y = PE_{BR,y} + PE_{EC,y}$$

Equation (4)

Parameter	Description	Unit	21-Jan-2022-31-Dec-2022	2023	Total
PE_y	Project emissions	tCO ₂ e	8,738.00	9,227.00	17,965.00
$PE_{BR,y}$	Project emissions from biogas recovery	tCO ₂ e	7,703.34	8,621.38	16,324.71
$PE_{EC,y}$	Project emissions from electricity consumption	tCO ₂ e	1,033.81	605.46	1639.27

$$PE_{BR,y} = (1 - CFE_{ww}) \times MEP_{ww,treatment,y} \times GWP_{CH4}$$

Equation (5)

Parameter	Description	Unit	21-Jan-2022-31-Dec-2022	2023	Total
$PE_{BR,y}$	Project emissions from biogas recovery	tCO ₂ e	7,703.34	8,621.38	16,324.71
CFE_{ww}	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems	/	0.9	0.9	/
$MEP_{ww,treatment,y}$	Methane emission potential of wastewater treatment systems equipped with biogas recovery system	t	2,751.19	3,079.06	5,830.26
GWP_{CH4}	Global Warming Potential for methane	tCO ₂ e/tCH ₄	28	28	/

$$MEP_{ww,treatment,y} = Q_{ww,y} \times B_{o,ww} \times UF_{PJ} \times \sum_k COD_{removed,PJ,k,y} \times MCF_{ww,treatment,PJ,k}$$

Equation (6)

Parameter	Description	Unit	21-Jan-2022-31-Dec-2022	2023	Total
-----------	-------------	------	-------------------------	------	-------

$MEP_{ww,treatment,y}$	Methane emission potential of wastewater treatment systems equipped with biogas recovery system	t	2,751.19	3,079.06	5,830.26
$COD_{removed,PJ,k,y}$	The chemical oxygen demand removed ¹² by the treatment system k of the project activity equipped with biogas recovery	t/m ³	0.042088	0.043733	/
$MCF_{ww,treatment,PJ,k}$	Methane correction factor for the project wastewater treatment system k equipped with biogas recovery equipment	/	0.8	0.8	/
UF_{PJ}	Model correction factor to account for model uncertainties	/	1.12	1.12	/
$Q_{ww,i,y}$	Volume of wastewater treated in baseline wastewater treatment system	m ³	294,923.28	315,417.25	610,340.53
$B_{o,ww}$	Methane producing capacity of the wastewater	kg CH ₄ /kg COD	0.25	0.25	/

$$PE_{EC,y} = \sum EC_{PJ,j,y} * EF_{EF,j,y} * (1 + TDL_{j,y})$$

Equation(7)

Parameter	Description	Unit	21-Jan-2022-31-Dec-2022	2023	Total
$PE_{EC,y}$	Project emissions from electricity consumption	tCO _{2e}	1,033.81	605.46	1639.27
$EC_{PJ,j,y}$	Quantity of electricity consumed by the	MWh	1.506	0.882	2.388

¹² Difference between the inflow COD and the outflow COD.

	project electricity consumption source j				
$EF_{EF,j,y}$	Emission factor for electricity generation for source j	tCO ₂ /MWh	0.57205	0.57205	/
$TDL_{j,y}$	Average technical transmission and distribution losses for providing electricity to source j	/	0.2	0.2	/

5.3 Leakage Emissions

There is no equipment transferred in the project, no leakage effects need to be accounted under AMS-III.H. and AMS-I.D.

5.4 GHG Emission Reductions and Carbon Dioxide Removals

Case 1: Ex post calculated baseline, project and leakage emissions based on actual monitored data for the project activity.

$$ER_{y,ex,post,1} = BE_{y,ex,post} - PE_{y,ex,post} - LE_{y,ex,post}$$

Equation (8)

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
21-Jan-2022 to 31-Dec-2022	66,334	8,738	0	57,596	0	57,596
01-Jan-2023 to 31-Dec-2023	75,253	9,227	0	66,026	0	66,026
Total	141,587	17,965	0	123,622	0	123,622

Case 2: The amount of biogas recovered and fuelled or flared (MD_y) during the crediting period, that is monitored ex post.

$$ER_{y,ex,post,2} = MD_y - PE_{power,y} - PE_{biomass,y} - LE_{y,ex,post}$$

Equation (9)

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
----------------	---	--	--	-------------------------------------	-----------------------------------	---------------------------------

21-Jan-2022 to 31-Dec-2022	67,127	862	0	66,265	0	66,265
01-Jan-2023 to 31-Dec-2023	88,379	505	0	87,874	0	87,874
Total	155,506	1,367	0	154,139	0	154,139

$$MD_y = BG_{burnt,y} \times W_{CH_4,y} \times D_{CH_4} \times FE \times GWP_{CH_4} \quad \text{Equation (10)}$$

Parameter	Description	Unit	21-Jan-2022-31-Dec-2022	2023	Total
MD _y	Methane captured and destroyed/gainfully used by the project activity	t CO ₂ e	67,127.00	88,379.00	155,506.00
BG _{burnt,y}	Biogas ¹³ combusted	m ³	5,136,216.00	6,765,816.00	11,902,032.00
w _{CH₄,y}	Methane content ¹³ of the biogas	%	65.23%	65.15%	/
D _{CH₄}	Density of methane at the temperature and pressure of the biogas	t/m ³	0.0007156	0.0007156	/
FE	Flare efficiency	%	100	100	/

$$PE_{power,y} = EG_{PJ,y} \times EF_{grid,y} \quad \text{Equation (11)}$$

Parameter	Description	Unit	21-Jan-2022-31-Dec-2022	2023	Total
PE _{power,y}	Emissions from electricity or fuel consumption	t CO ₂ e	862.00	505.00	1,367.00
EC _{PJ,y}	Quantity of electricity consumed by the project electricity consumption source j	MWh	1.506	0.882	2.388
EF _{grid,y}	Combined margin CO ₂ emission factor for grid connected power generation	tCO ₂ /MWh	0.57205	0.57205	/

¹³ Biogas volume and methane content measurements shall be on the same basis (wet or dry).

The water treatment system remains unchanged, and only methane capture system was installed. Therefore, $PE_{biomass,y}$ is equal to 0.

$$ER_{y,ex,post} = \min((BE_{y,ex,post} - PE_{y,ex,post} - LE_{y,ex,post})(MD_y - PE_{power,y} - PE_{biomass,y} - LE_{y,ex,post})) \quad \text{Equation (12)}$$

According to the Equation (12), the ex post emission reductions shall be based on the lowest value of the following: (1) Case1: Ex post calculated baseline, project and leakage emissions based on actual monitored data for the project activity. (b) Case2: The amount of biogas recovered and fuelled or flared (MD_y) during the crediting period, that is monitored ex post. The data shows that $ER_{y,ex,post,1}$ is less than $ER_{y,ex,post,2}$, so case 1 applies. $ER_{y,ex,post} = ER_{y,ex,post,1} = BE_{y,ex,post} - PE_{y,ex,post} - LE_{y,ex,post}$ Equation (13)

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
21-Jan-2022 to 31-Dec-2022	66,334	8,738	0	57,596	0	57,596
01-Jan-2023 to 31-Dec-2023	75,253	9,227	0	66,026	0	66,026
Total	141,587	17,965	0	123,622	0	123,622

The monitoring period covers from 21-Jan-2022 to 31-Dec-2023, with totally 710 days. Based on the registered VCS PD, the amount of estimated emission reductions for this monitoring period would be 62,677+66,310=128,987tCO₂e. The actual emission reductions in this monitoring period are 123,622tCO₂e, which is 4.16% less than the estimation in the registered VCS PD.

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
21-Jan-2022 to 31-Dec-2022	62,677	57,596	-8.11%	The values in PD are derived from the actual project operation data in 2022 and the EIA report. All the data in this monitoring period are measured. 1) The amount of net electricity supplied to the grid

				used in PD is 11,198 MWh (21-Jan-2022 to 31-Dec-2022), but the actual amount is 8,950.628 MWh (21-Jan-2022 to 31-Dec-2022). 2) As 2022 is the first year of the project operation, the project operation management experience is insufficient. In particular, in the first quarter of 2022, less amount of net electricity was supplied to the grid.
01-Jan-2023 to 31-Dec-2023	66,310	66,026	-0.43%	For biogas power project, with many external factors affecting the amount and quality of biogas generated, including climate, quantity of leachate, chemical oxygen demand removed by the treatment system, leachate treatment efficiency, gas collection efficiency and technology etc., it is reasonable that the measured value deviates from the predicted value.
Total	128,987	123,622	-4.16%	For biogas power project, with many external factors affecting the amount and quality of biogas generated, including climate, quantity of leachate, chemical oxygen demand removed by the treatment system, leachate treatment efficiency, gas collection efficiency and technology etc., it is reasonable that the measured value deviates from the predicted value. Comparing to the estimated emission reductions, the mainly reasons of the difference are for that: The values in PD are derived from the actual project operation data in 2022 and the EIA report. All the data in this monitoring period are measured. 1) The amount of net electricity supplied to the grid used in PD is 11,198 MWh (21-Jan-2022 to 31-Dec-2022),

				but the actual amount is 8,950.628 MWh (21-Jan-2022 to 31-Dec-2022). 2) As 2022 is the first year of the project operation, the project operation management experience is insufficient. In particular, in the first quarter of 2022, less amount of net electricity was supplied to the grid. Therefore, the decrease 4.16% is reasonable and acceptable for biogas power project.
--	--	--	--	--

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

The monitoring report does not involve commercially sensitive information.

Section	Information	Justification
/	/	/

APPENDIX 2: <SUPPORTING EVIDENCE>

Supporting evidence for contribution SDG 13

Evidence: Emission reductions achieved during the First Monitoring Period

Emission reductions are calculated as follows:

$$ER_{y,ex,post} = \min((BE_{y,ex,post} - PE_{y,ex,post} - LE_{y,ex,post})(MD_y - PE_{power,y} - PE_{biomass,y} - LE_{y,ex,post}))$$

Equation (12)

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
21-Jan-2022 to 31-Dec-2022	66,334	8,738	0	57,596	0	57,596
01-Jan-2023 to 31-Dec-2023	75,253	9,227	0	66,026	0	66,026
Total	141,587	17,965	0	123,622	0	123,622

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
21-Jan-2022 to 31-Dec-2022	62,677	57,596	-8.11%	The values in PD are derived from the actual project operation data in 2022 and the EIA report. All the data in this monitoring period are measured. 1) The amount of net electricity supplied to the grid used in PD is 11,198 MWh (21-Jan-2022 to 31-Dec-2022), but the actual amount is 8,950.628 MWh (21-Jan-2022 to 31-Dec-2022). 2) As 2022 is the first year of the project operation, the project operation management experience is insufficient. In particular, in the first quarter of 2022, less amount of net

				electricity was supplied to the grid.
01-Jan-2023 to 31-Dec-2023	66,310	66,026	-0.43%	For biogas power project, with many external factors affecting the amount and quality of biogas generated, including climate, quantity of leachate, chemical oxygen demand removed by the treatment system, leachate treatment efficiency, gas collection efficiency and technology etc., it is reasonable that the measured value deviates from the predicted value.
Total	128,987	123,622	-4.16%	For biogas power project, with many external factors affecting the amount and quality of biogas generated, including climate, quantity of leachate, chemical oxygen demand removed by the treatment system, leachate treatment efficiency, gas collection efficiency and technology etc., it is reasonable that the measured value deviates from the predicted value. Comparing to the estimated emission reductions, the mainly reasons of the difference are for that: The values in PD are derived from the actual project operation data in 2022 and the EIA report. All the data in this monitoring period are measured. 1) The amount of net electricity supplied to the grid used in PD is 11,198 MWh (21-Jan-2022 to 31-Dec-2022), but the actual amount is 8,950.628 MWh (21-Jan-2022 to 31-Dec-2022). 2) As 2022 is the first year of the project operation, the project operation management experience is insufficient. In particular, in the first quarter of 2022, less amount of net electricity was supplied to the grid.

				Therefore, the decrease 4.16% is reasonable and acceptable for biogas power project.
--	--	--	--	--

Source: Section 5.4 of the First Monitoring Report of VCS Project 3851.

Supporting evidence for contribution SDG 8

Evidence: Total number of job opportunities: Zhongmu Branch company of Henan BCCY Environmental Energy Co., Ltd

员工花名册					
序号	所属电厂	姓名	性别	职位	人员状态
1	中牟电厂	陶金龙	男	厂长	在职
2	中牟电厂	徐普州	男	运营主管	在职
3	中牟电厂	尚建坡	男	运维员	在职
4	中牟电厂	李兆华	男	检修员	在职
5	中牟电厂	李玮	男	运维员	在职
6	中牟电厂	张建鑫	男	运维员	在职
7	中牟电厂	陈钰帛	女	会计	在职
8	中牟电厂	林淑花	女	厨师	在职
9	中牟电厂	赵海洋	男	运维员	在职
10	中牟电厂	潘国杰	男	收集员	在职

Source: Employee file of Zhongmu Branch company of Henan BCCY Environmental Energy Co., Ltd

Supporting evidence for contribution SDG 7

Evidence: Affordable and Clean Energy: Zhongmu Branch company of Henan BCCY Environmental Energy Co., Ltd

MONTH	EG _{PJ, facility, y}	EC _{PJ, y}
	Monthly net quantity of electricity produced using biogas	Monthly electricity from grid to project
	kWh	kWh
21-Jan-2022-31-Jan-2022	41,698.00	92.00
01-Feb-2022-28-Feb-2022	111,181.00	-
01-Mar-2022-31-Mar-2022	123,140.00	-
01-Apr-2022-31-Apr-2022	653,616.00	258.00
01-May-2022-31-May-2022	1,018,137.00	275.00
01-Jun-2022-31-Jun-2022	887,451.00	181.00
21-Jan-2022-31-Jan-2022	1,654,245.00	298.00
01-Jul-2022-31-Jul-2022	1,579,188.00	242.00
01-Aug-2022-31-Aug-2022	1,375,424.00	-
01-Sep-2022-31-Sep-2022	822,498.00	-
01-Oct-2022-31-Oct-2022	529,625.00	160.00
01-Nov-2022-31-Nov-2022	154,425.00	-
subtotal	8,950,628.00	1,506.00
01-Jan-2023-31-Jan-2023	251,156.00	-
01-Feb-2023-28-Feb-2023	679,680.00	-
01-Mar-2023-31-Mar-2023	879,665.00	-
01-Apr-2023-31-Apr-2023	844,862.00	-
01-May-2023-31-May-2023	1,119,464.00	162.00
01-Jun-2023-31-Jun-2023	1,160,254.00	263.00
01-Jul-2023-31-Jul-2023	1,658,630.00	177.00
01-Aug-2023-31-Aug-2023	1,259,868.00	201.00

01-Sep-2023- 31-Sep-2023	1,155,841.00	-
01-Oct-2023- 31-Oct-2023	1,143,197.00	79.00
01-Nov-2023- 31-Nov-2023	1,001,146.00	-
01-Dec-2023- 31-Dec-2023	635,746.00	-
subtotal	11,789,509.00	882.00
Total	20,740,137.00	2,388.00

Source: Daily operation records from Zhongmu Branch company of Henan BCCY Environmental Energy Co., Ltd and receipt of electricity purchase and sales from grid company.